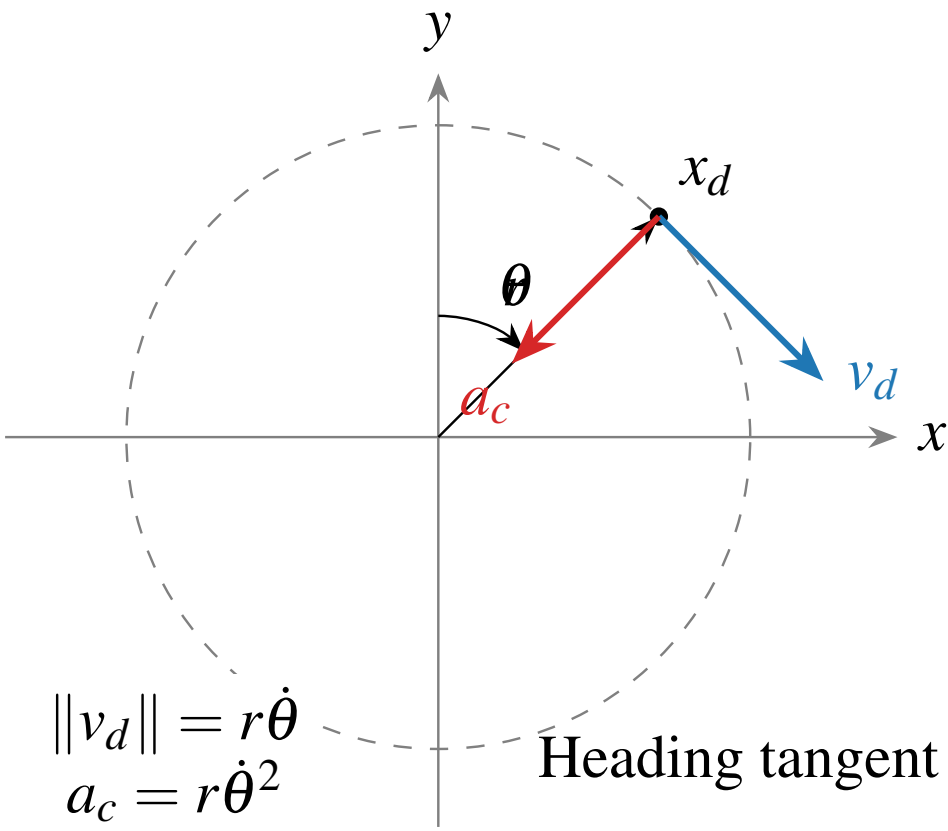
(a) Reference motion in the horizontal plane

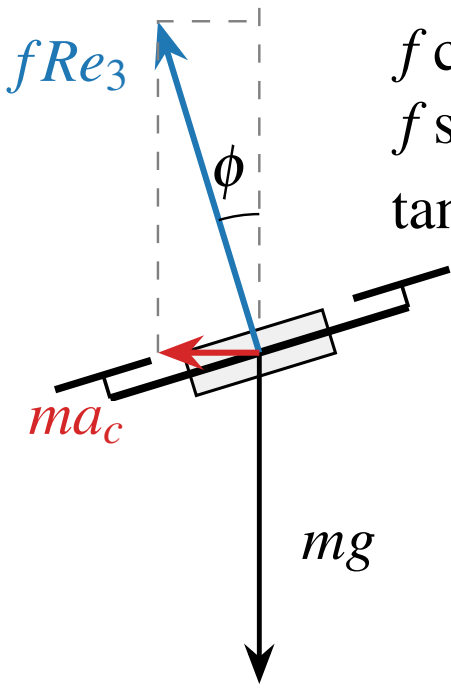


$\|v_d\| = r\dot{\theta}$

$a_c = r\dot{\theta}^2$

Heading tangent to path

(b) Thrust during a level coordinated turn



$f\cos\phi = mg$

$f\sin\phi = ma_c$

$\tan\phi = a_c/g$

$r = 5$ m, $\quad v = 2.5$ m/s

$a_c = 1.25$ m/s^2, $\quad \phi = 7.26°$